An analysis of data on drive_secondary in Scotland

Group31: Yaxin Xu, Zixing Wang, Wanyu Lin, Yirui Zhao

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Summary

In this report, we investigate the time spent driving to secondary schools in different regions of Scotland. We applied Welch's t-test and an F-test to compare the drive time in Glasgow and Edinburgh, Pearson correlation and simple linear regression for modeling drive time, and Spearman correlation, two-way ANOVA with interaction test, Tukey HSD, and multiple linear regression with interaction terms for testing spatial heterogeneity of educational accessibility. These results show that Edinburgh has both poorer average accessibility and more unequal school proximity, and drive time to retail centres emerges as the strongest linear predictor of drive time to secondary school. Furthermore, the income–drive time relationship exhibits an overall "poorer–closer" trend but differs by region: Edinburgh exhibits a significant negative linear association, Glasgow shows no linear association, and the Rest of Scotland falls in between, demonstrating that national-level correlations conceal local differences.

Introduction

This study investigates driving time to secondary schools in Scotland through three independent yet progressive perspectives. First, we compare the mean driving time between Glasgow and Edinburgh. Second, we compute Pearson correlation coefficients between drive_secondary (average driving time in minutes from a data zone to a secondary school) and four other variables, select the variable with the highest correlation, and fit a simple linear regression model. Third, we explore the relationship between driving time and area-level wealth, test whether this relationship is consistent across Glasgow, Edinburgh, and the rest of Scotland, and further assess whether the overall association observed in the full sample is driven by the characteristics of major cities. In Scotland, differences in driving time to secondary schools across regions may reflect imbalances in area-level wealth, housing overcrowding, and the accessibility of both car and public transport. The data used in this study come from 400 Scottish data zones – 100 from Glasgow City, 100 from the City of Edinburgh, and the remaining 200 from the rest of Scotland. In addition to the core variable drive_secondary, four representative indicators were selected: average driving time to a retail centre (drive_retail), public transport travel time to a retail centre (PT_retail), the proportion of people living in overcrowded households (overcrowded_rate), and the proportion of income-deprived people (Income_rate), defined as the percentage of people in a data zone who are income deprived according to benefit-based measures. By evaluating driving time to secondary schools and its associations with these indicators, this study aims to provide empirical evidence on inequalities in educational accessibility, supporting improved school placement and transport infrastructure.

Methods

All analyses were performed using R. In Section 1, to compare drive_secondary between Glasgow and Edinburgh, we produced side-by-side boxplots and separate histograms, calculated sample means and variances, derived 95% confidence intervals for means from the t-distribution and for variances from the chi-squared distribution, and used a Welch's two-sample t-test for means and an F-test for variances ($\alpha = 0.05$). In Section 2, to identify the variable most linearly associated with drive_secondary in the full sample, we computed Pearson correlations between drive_secondary and four variables (drive_retail, PT_retail, overcrowded_rate, Income_rate), selected the highest-correlation predictor, fitted a simple linear regression, and examined a residual plot to assess linearity, constant variance, and independence. In Section 3, to test whether the relationship between income deprivation and driving time varies by region, we first grouped Income_rate into quartiles (Q1 richest to Q4 poorest), simplified council area into three regions (Glasgow, Edinburgh, Rest of Scotland), performed a two-way ANOVA with drive_secondary as the dependent variable and income quartile and region as independent factors, and used Tukey's HSD test for post-hoc comparisons. Then, to quantify the moderating effect of region, we built a multiple linear regression model including the continuous Income_rate, region (with Glasgow as baseline), and their interaction term. All continuous variables were Z-scaled before analysis to allow coefficient interpretation in standard deviation units. This regression directly tests whether the effect of income deprivation on driving time differs across regions and provides effect sizes.

Results

Section 1: Comparing drive time to secondary schools between the cities of Glasgow and Edinburgh

Figure 1 presents side-by-side box-plots of drive time to secondary schools for Glasgow and Edinburgh. The median drive time is slightly higher in Edinburgh than in Glasgow, and the interquartile range (IQR) is notably wider for Edinburgh, indicating greater variability among its data zones.

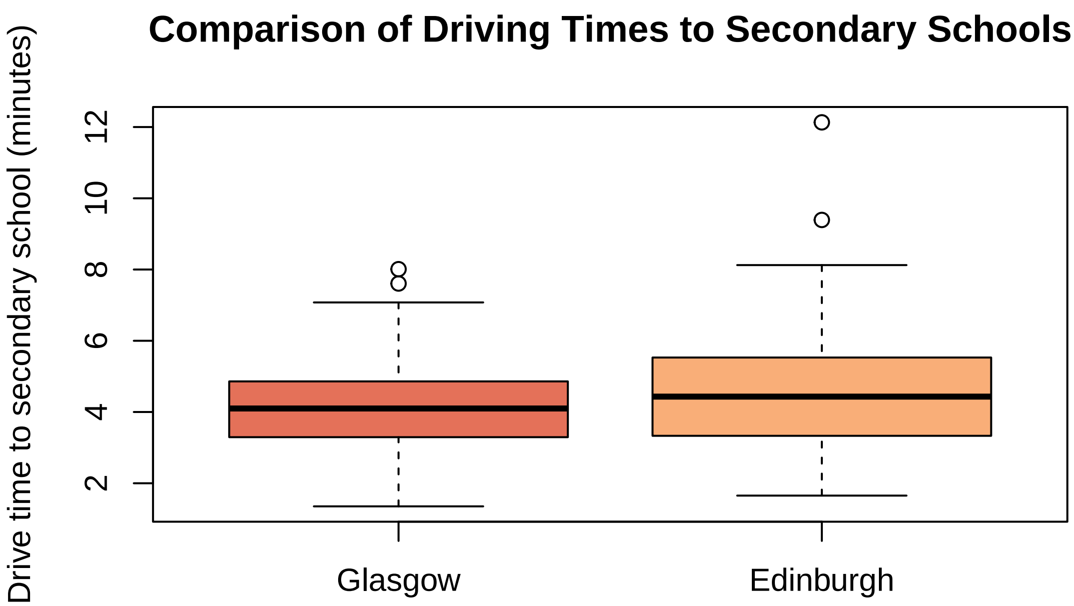


Figure 1: Side-by-side boxplots of drive time to secondary schools for Glasgow and Edinburgh.

Figure 2 reinforces this pattern. The Glasgow histogram is roughly symmetric, with most observations concentrated between 3 and 5 minutes. In contrast, the Edinburgh histogram exhibits a longer right tail, implying a small number of zones with considerably longer drive times (exceeding 8 minutes). This means that Edinburgh has a few outlier areas where car access to secondary schools is notably poor.

Summary statistics in Table 1 confirm our finding that Glasgow has a lower mean (4.15 minutes) and smaller variance (1.66) compared to Edinburgh (mean = 4.64, variance = 2.82). The 95% confidence intervals for the means and variances are also provided.

Table 1: Summary statistics and 95% confidence intervals for drive_secondary in Glasgow and Edinburgh

Statistic	Glasgow	Edinburgh
Mean	4.15	4.64
Variance	1.66	2.82
95% CI for Mean	(3.89, 4.4)	(4.31, 4.97)
95% CI Variance	(1.28, 2.24)	(2.17, 3.81)

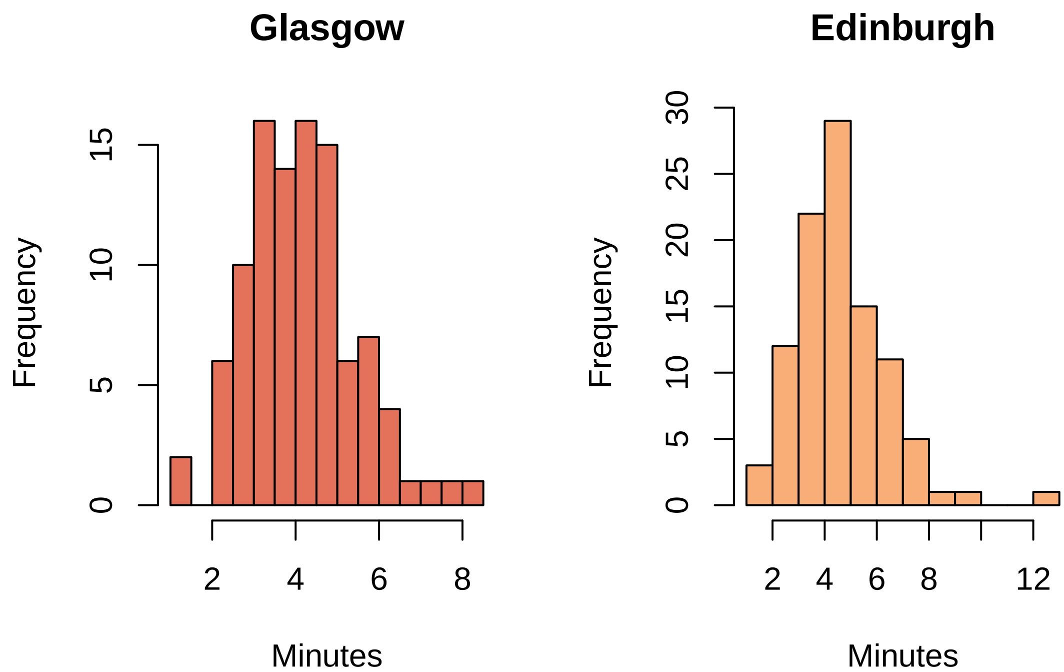


Figure 2: Histograms of drive time to secondary schools for Glasgow and Edinburgh.

Hypothesis tests:

Difference in means (Welch two-sample t-test): The test yielded a t-statistic of approximately −2.33 (degrees of freedom 185.56) and a p-value of 0.0208. Since this p-value is less than 0.05, we conclude there is evidence against the null hypothesis of equal means, indicating that the average drive time to a secondary school differs between Glasgow and Edinburgh – with Edinburgh's mean being about half a minute longer.(Although the absolute difference is modest, the consistency across 100 zones makes it statistically meaningful.)

Difference in variances (F-test): The F-test for equality of variances gave a p-value of approximately 0.0089 ($F \approx 0.589, df_1 = df_2 = 99$), which is below 0.01, so we reject the null hypothesis that the variances are equal with strong evidence. The results show that Edinburgh's drive times are not only higher on average but also significantly more variable, indicating greater inequality in secondary school accessibility across its data zones.

Section 2: Investigating the relationship between drive time to secondary schools and other variables

In this section, we use all 400 observations (including Glasgow, Edinburgh, and other council areas) to explore linear relationships among five variables: drive_secondary (dependent variable of interest), drive_retail, PT_retail, overcrowded_rate, and Income_rate. The first two are drive-time measures; PT_retail is public transport travel time to a retail centre; the latter two are indicators of housing overcrowding and income deprivation.

Pairwise scatterplots (Figure 3) show positive linear associations between drive-time and public transport variables, while deprivation indicators exhibit negative correlations with accessibility measures. No obvious non-linear patterns are observed.

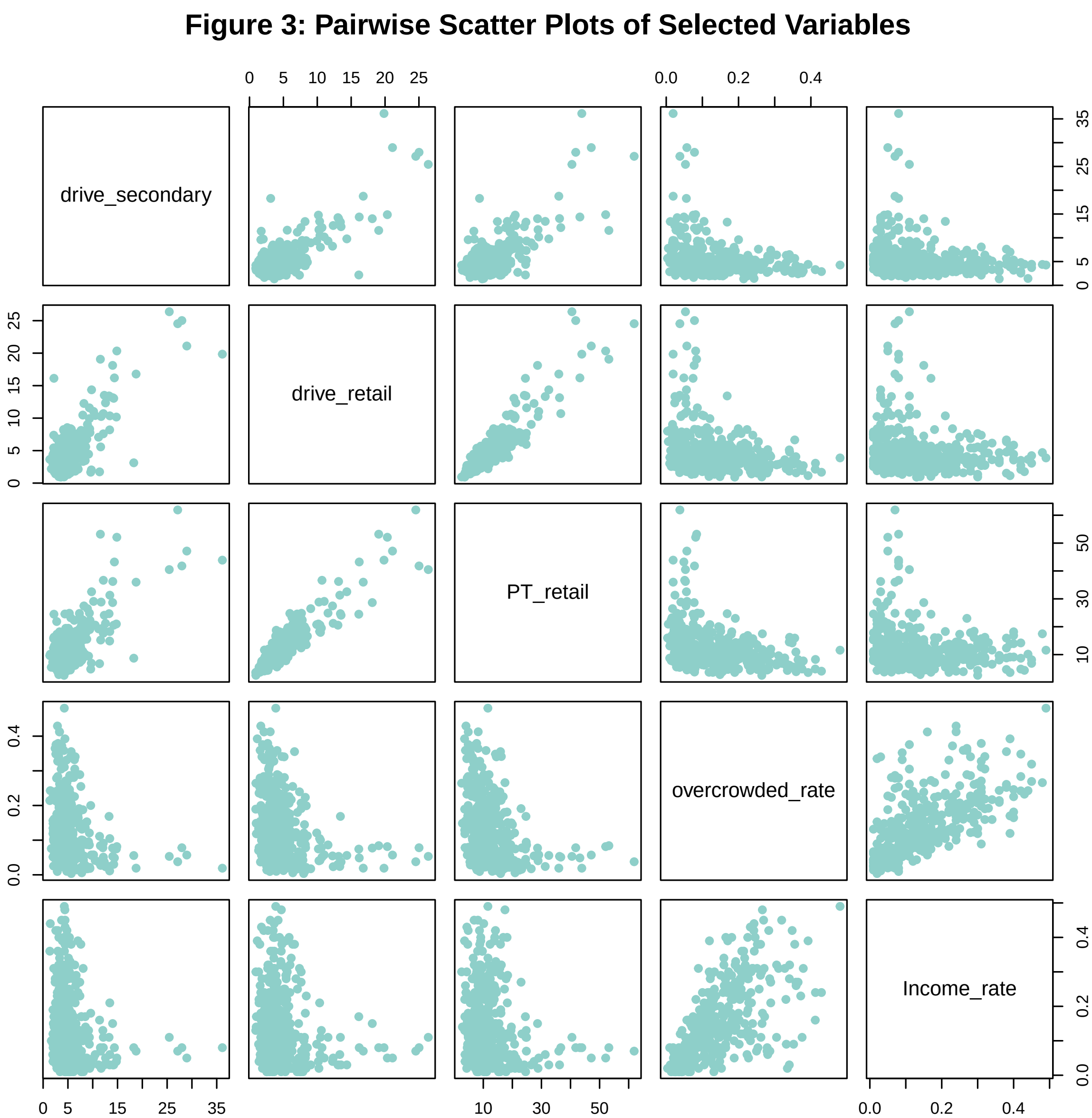


Figure 3: Pairwise Scatter Plots of Selected Variables

Pearson correlation coefficients (Table 2) reveal that drive_secondary is most strongly correlated with drive_retail (r = 0.777, p < 0.001). It is also moderately positively correlated with PT_retail (r = 0.719, p < 0.001), and weakly negatively correlated with overcrowded_rate (r = –0.272, p = 3.25×10) and Income_rate (r = –0.214, p = 1.60×10). Based on the highest absolute correlation, we select drive_retail as the independent variable data2 for subsequent regression analysis. The p-values corresponding to these Pearson correlation coefficients are shown in Table 3.

Table 2: Pearson Correlation Coefficients

	drive_secondary	drive_retail	PT_retail	overcrowded_rate	Income_rate
drive_secondary	1.000	0.777	0.719	-0.272	-0.214
drive_retail	0.777	1.000	0.913	-0.295	-0.199
PT_retail	0.719	0.913	1.000	-0.326	-0.208
overcrowded_rate	-0.272	-0.295	-0.326	1.000	0.663
Income_rate	-0.214	-0.199	-0.208	0.663	1.000

Table 3: P-values for Correlations

NA	6.1927e-82	7.1309e-65	3.2466e-08	1.6012e-05
6.1927e-82	NA	2.1984e-157	1.8516e-09	6.2057e-05
7.1309e-65	2.1984e-157	NA	2.2204e-11	2.8273e-05
3.2466e-08	1.8516e-09	2.2204e-11	NA	6.1194e-52
1.6012e-05	6.2057e-05	2.8273e-05	6.1194e-52	NA

Simple linear regression:
We regress drive_secondary on drive_retail, obtaining the following linear equation: drive_secondary = 1.4643 + 0.8317 × drive_retail. Both coefficients are highly significant (p < 0.001). The coefficient of determination $R^2 = 0.6035$ (the square of the correlation r = 0.777) indicates that approximately 60.4% of the variation in drive_secondary is explained by drive_retail.

Scatter plot with regression line (Figure 4) displays a clear positive trend: for each additional minute of drive time to a retail centre, the predicted drive time to a secondary school increases by about 0.83 minutes. The fitted line agrees well with the majority of observations.

Residual plot (Figure 5) shows residuals against fitted values. The residuals are randomly scattered around zero, with no obvious funnel shape or systematic curvature, supporting the assumptions of linearity and homoscedasticity. A few points with large fitted values (above 15 minutes) exhibit slightly larger residuals, but no severe violations of model assumptions are evident.

In summary, drive_retail is a strong linear predictor of drive_secondary, and the regression assumptions appear reasonably satisfied. This suggests that accessibility to retail and secondary schools is closely linked in Scotland, likely reflecting spatial clustering of services.

Section 3: From National Trends to Regional Moderation: Analyzing Educational Accessibility

3.1 Overall Pattern: The Preliminary Relationship between Income Deprivation and Drive Time to Secondary School in Scotland

Before delving deeper, we first performed a macroscopic description of the full sample (N = 400) to determine whether there is an overall association between income deprivation and drive time to secondary school across the whole of Scotland. For this purpose, we calculated the Spearman rank correlation coefficient between drive_secondary and Income_rate (income deprivation rate) and produced a boxplot grouped by income deprivation quartiles.

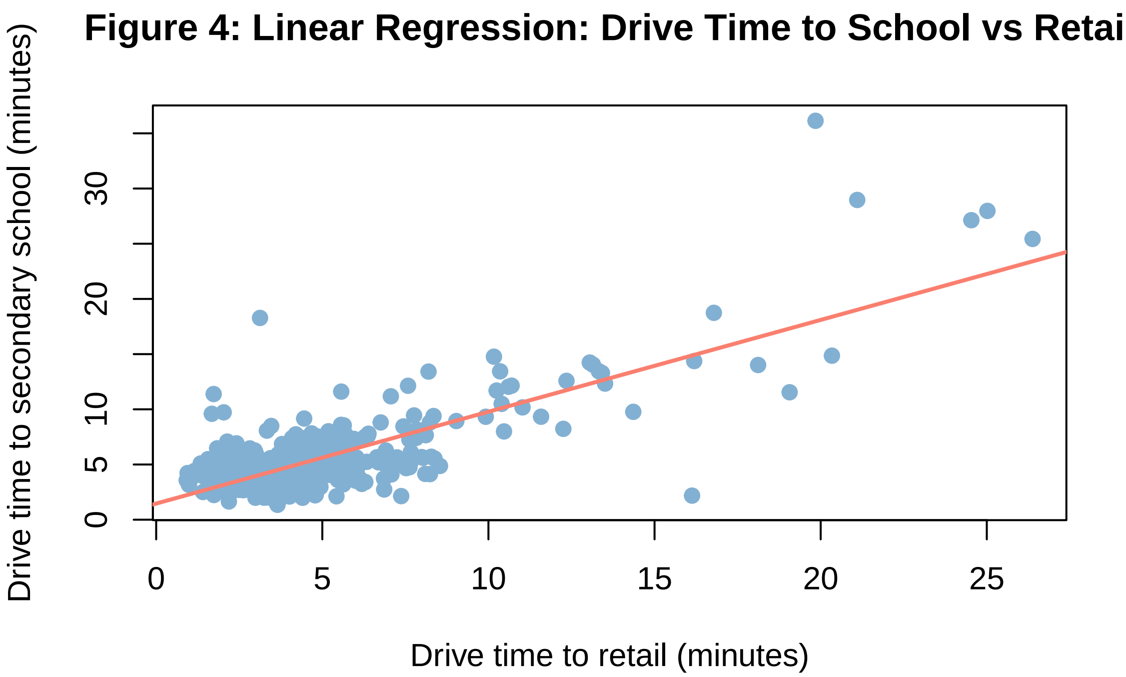


Figure 4: Linear Regression: Drive Time to School vs Retail

First, the strength and direction of the monotonic relationship between the income deprivation rate (Income_rate) and average drive time to a secondary school (drive_secondary) were quantified using Spearman's rank correlation coefficient. The non-parametric Spearman coefficient was precisely chosen over the standard Pearson coefficient, whose reliance on ranks makes it robust to extreme values. This is especially important here: some remote data zones in the Highlands and Islands may have unusually long drive times that could heavily distort a Pearson-based analysis.

The calculation yields a Spearman coefficient $\rho = -0.2364 (p < 0.001)$, indicating a statistically significant but modest negative monotonic association at the national scale. In plain terms, data zones located closer to a secondary school tend to suffer from somewhat higher levels of income deprivation. While counter-intuitive at first glance — one might expect neighbourhoods near good schools to be more affluent — this signal provides the puzzle that the rest of the section sets out to unpack.

To further visualise the structure behind this association, drive_secondary was divided into four groups according to the quartiles of Income_rate (Q1: least deprived; Q4: most deprived), and a side-by-side boxplot was constructed. This grouping strategy neatly sidesteps the overplotting problem that can plague scatterplots of highly dispersed variables, making distributional differences immediately visible. The figure reveals a clear descending gradient: as income deprivation intensifies, the median drive time to a secondary school systematically shortens. Specifically, the median drive time drops from 5.25 minutes in the least deprived quartile (Q1) to 4.24 minutes in the most deprived quartile (Q4). At the same time, the spread of the data narrows considerably — the interquartile range (IQR) shrinks from 3.59 minutes (Q1: 3.98-7.57) to just 1.6 minute (Q4: 3.37–4.97) — suggesting that the most deprived data zones are not only physically closer to secondary schools but also more homogeneous in their geographic accessibility.

However, despite the systematic downward shift of the box bodies, there remains substantial overlap between the upper and lower fences across all quartile groups. Although the central tendency of "poorer means closer" is statistically clear, it is insufficient to reach a common law. In addition, such a pattern strongly hints at marked regional heterogeneity or the presence of third variables that distort the relationship in certain

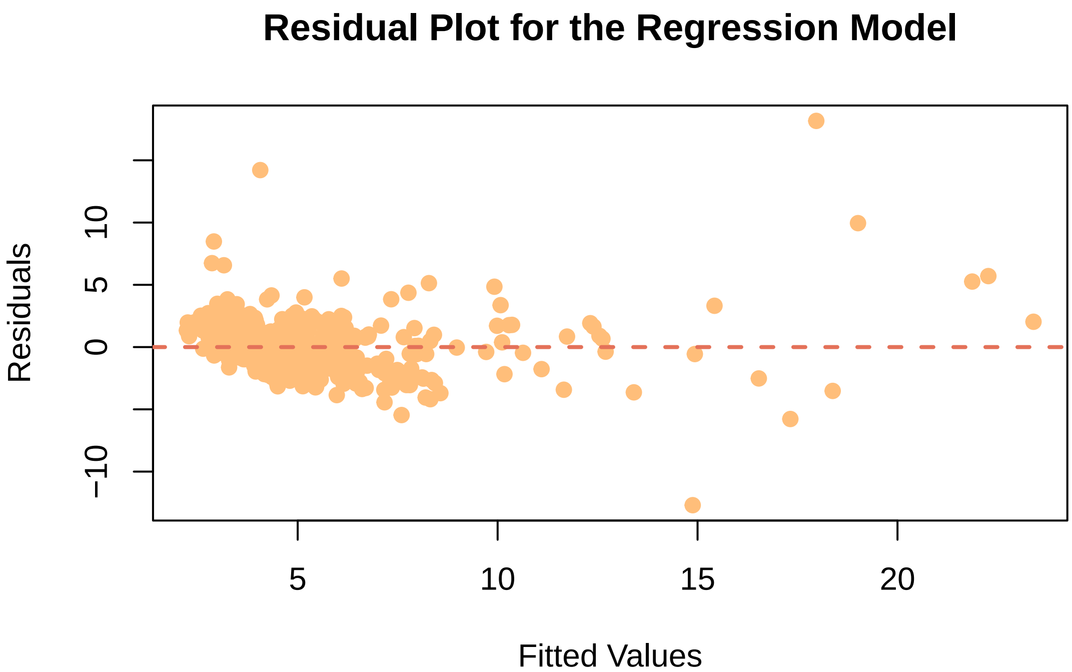


Figure 5: Residual Plot for the Regression Model

pockets of the data, which aligns well with the modest magnitude of the Spearman coefficient ρ of -0.24: This naturally motivates the next analytical step in the section 3.2, which breaks the data down by region — Glasgow, Edinburgh and elsewhere — to see whether the overall negative association holds with the same direction and strength within these areas.

3.2 Regional Divergence in the Relationship between Income Deprivation and Educational Accessibility

3.2.1 Why This Further Investigation?

Section 3.1 provided an overall judgement through a single national-level Spearman correlation coefficient. However, the validity of such a summary measure rests on an implicit assumption: that the relationship between income deprivation and drive time to secondary school is broadly homogeneous across all data zones in Scotland. Once this relationship exhibit different slopes in different regions, a single national ρ ceases to be a reliable summary.

The core task of this section, therefore, shifts from "measuring association" to "detecting variation": does the relationship change depending on region?

Our analytical strategy operates at two levels. At the descriptive level, we apply scatter plots, grouped box plots and region-specific correlation coefficients to visually and numerically sense the association patterns within each region. At the inferential level, we employ a two-way ANOVA method, with Income_rate as the dependent variable, and drive_secondary and Region as the two factors, together with their interaction term. Compared to the alternative of simply estimating three separate regression lines and presenting them side by side, the two-way ANOVA offers a distinct advantage, since its interaction test provides a direct p-value for whether the differences among the slopes themselves are statistically significant. If the interaction proves significant, region-specific simple linear regressions are then estimated to characterise the shape of the

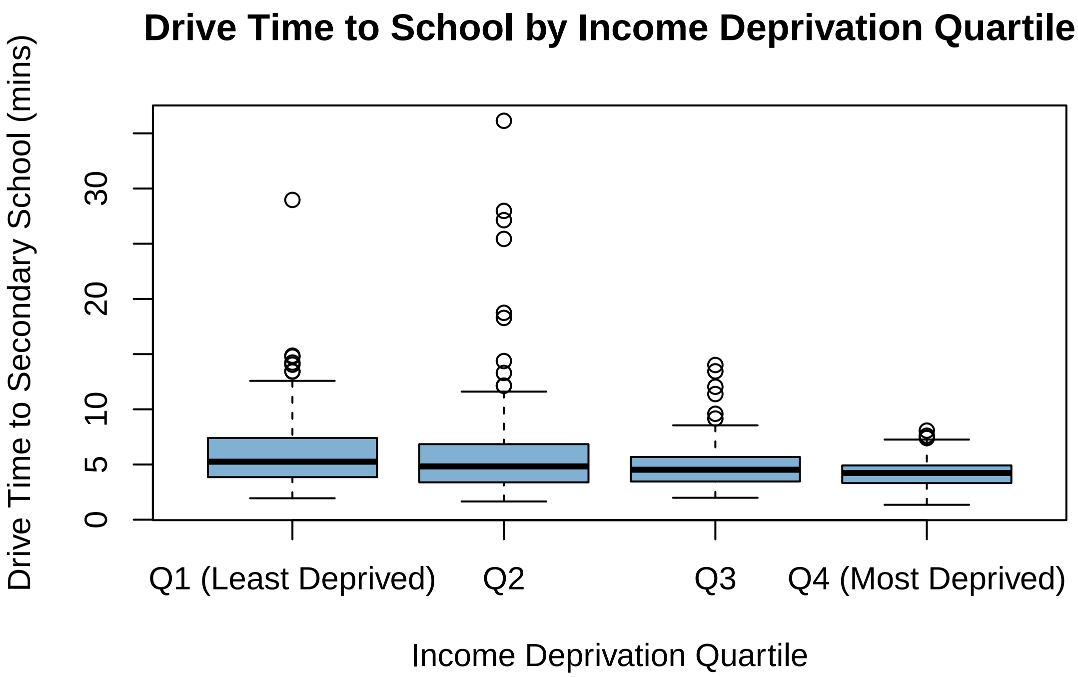


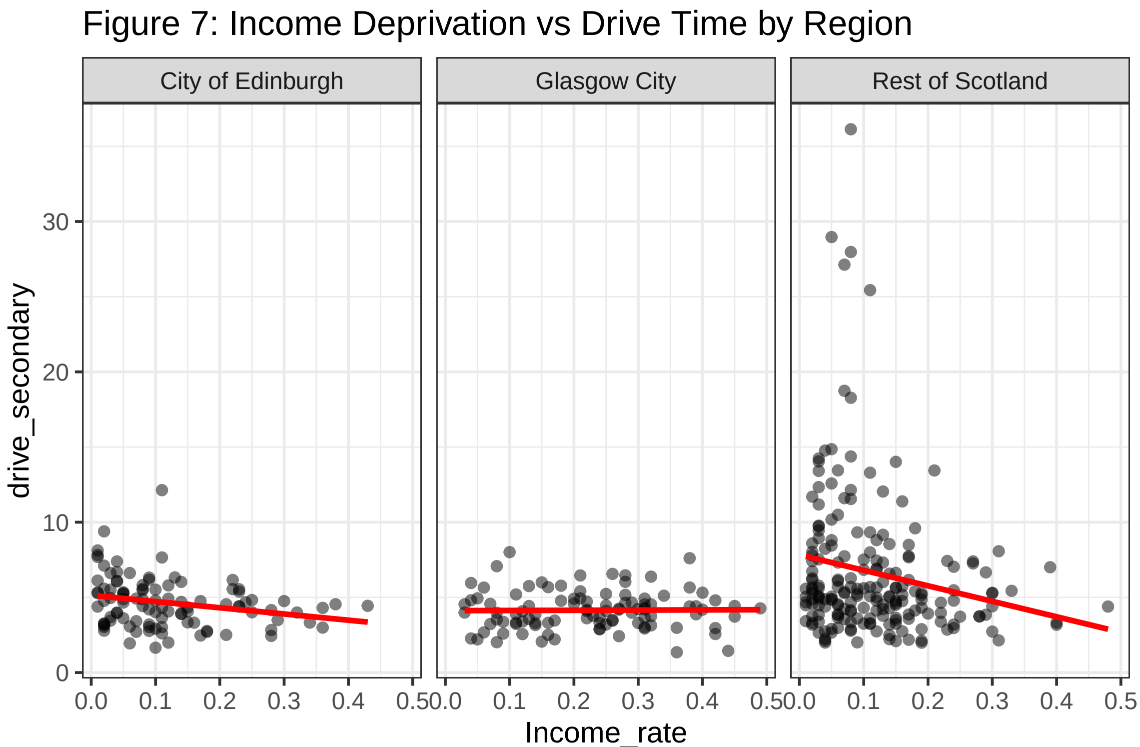
Figure 6: Boxplot of Drive Time by Income Deprivation Quartile

divergence, supplemented by a Tukey HSD post-hoc test to compare the absolute levels of drive time across the three regions. In this way, the section seeks a balance between descriptive intuitiveness and inferential rigour.

3.2.2 Visual Exploration: Faceted Scatterplots and Grouped Bloxplots

To obtain an initial visual impression, we approached the data from two complementary views.

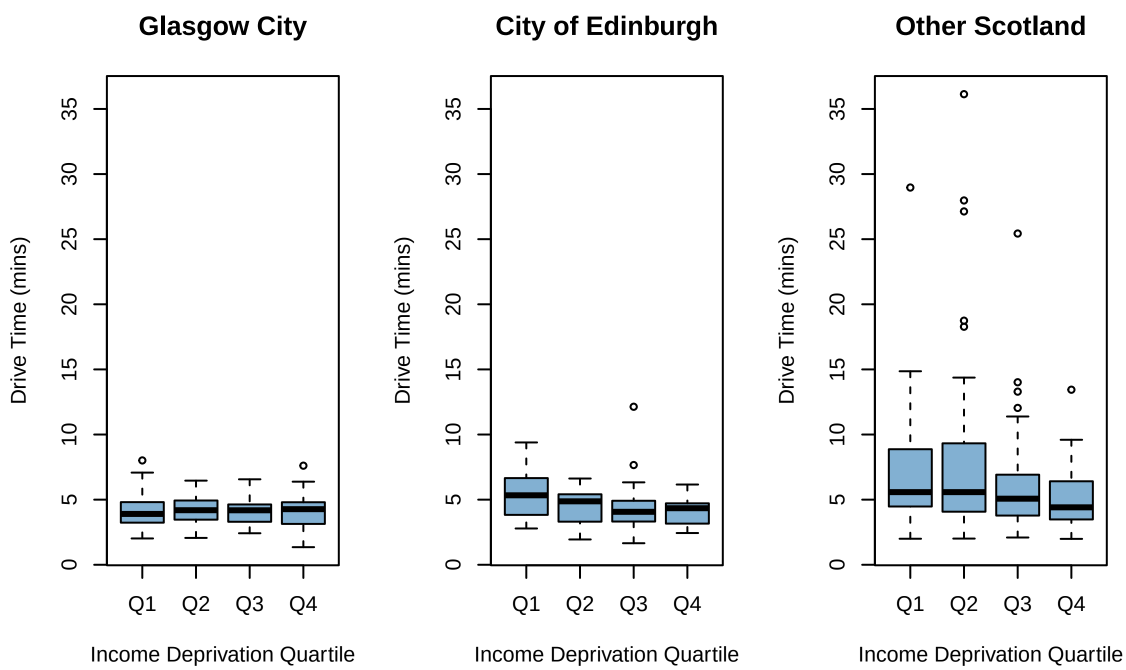
First, we produce scatterplots of `Income_rate` against `drive_secondary` separately for each of the three region groups, with linear trend lines overlaid.



The scattered plots in Figure 7 reveal clear regional differences. In Glasgow, the points cluster tightly at short drive times, and the trend line shows a distinct negative slope — higher deprivation is concentrated where drive times are lowest. In Edinburgh, the trend line is essentially flat, with deprived data zones scattered across the available range of drive times. The Rest of Scotland panel spans much wider drive times and displays only a very weak downward tilt amid substantial scatter.

Second, we grouped drive_secondary simultaneously by Region and income deprivation quartile (Q1 = least deprived, Q4 = most deprived) and produced a grouped boxplot.

Figure 8: Drive Time to Secondary School by Region & Income Deprivation



The boxplot brings out structural details that the scatterplot alone might obscure. In Glasgow, we observe a clear downward staircase of the median, insisting that beyond the most affluent group, drive time essentially stops varying with deprivation. Edinburgh presents a "two-tier" picture — Q1 stands apart with the longest median drive time, while Q2 through Q4 all sit at a little lower drive time. The Rest of Scotland exhibits a more complex, inverted-U-shaped fluctuation across quartiles, but an overall downward trend is still visible.

3.2.3 Correlation Coefficients by Region

To quantify the strength of association within each region, we computed Spearman correlation coefficients between Income_rate and drive_secondary separately for the three groups. The results are summarised below.

Table 4: Spearman correlation between drive_secondary and Income_rate by region

Region	rho	p_value
Edinburgh	-0.2826	4.387e-03
Glasgow	0.0465	6.457e-01
Rest of Scotland	-0.1996	4.607e-03

The Rest of Scotland's $\rho = -0.20 (p = 0.005)$ captures the overall downward trend visible in the boxplot. Edinburgh's $\rho = -0.28 (p = 0.004)$ confirms a mild but significant negative monotonic relationship — though the boxplot clarifies that this is driven primarily by the Q1-versus-others contrast rather than a smooth gradient.

Glasgow's near-zero $\rho = 0.047 (p = 0.646)$ initially surprises, given the negative scatterplot slope and the boxplot staircase. But the tension is instructive. Glasgow's drive times are compressed into a narrow 1–7 minute window — range restriction alone depresses the coefficient. More substantively, with Q2–Q4 medians essentially flat, there genuinely is no monotonic trend across most of the sample. The correlation correctly reports this absence, while the boxplot captures the single group contrast with Q1 that the continuous coefficient misses. Both tools are needed.

3.2.4 Formal Interaction Test: Two-Way ANOVA

To test the interaction formally, we fitted a two-way ANOVA.

Table 5: Two-way ANOVA results: drive_secondary ~ Income_quartile * Region

Source	Df	Sum.Sq	Mean.Sq	F.value	p.value
Income_quartile	3	304.53	101.51	8.12	2.95e-05
Region	2	387.75	193.88	15.51	3.31e-07
Income_quartile:Region	6	151.41	25.24	2.02	6.22e-02
Residuals	388	4850.37	12.50	NA	NA

We interpret the three rows as follows.

3.2.5 First Level: Regional Differences — Does Drive Time Differ by Where You Live?

This row of the ANOVA tests the Region main effect: regardless of income, do average drive times differ across Glasgow, Edinburgh, and the Rest of Scotland?

The region main effect is highly significant ($p<0.001$). This tells us that where a data zone is located matters for how far its residents must travel to a secondary school. Two data zones with identical deprivation levels can face very different drive times simply because one is urban and the other rural. The Tukey HSD post-hoc test fills in the detail.

Table 6: Tukey HSD post-hoc test for Region (pairwise comparisons)

	Comparison	diff	lwr	upr	p_adj
City of Edinburgh-Glasgow City	City of Edinburgh-Glasgow City	-0.241	-1.417	0.935	8.80e-01
Rest of Scotland-Glasgow City	Rest of Scotland-Glasgow City	1.778	0.759	2.796	1.46e-04
Rest of Scotland-City of Edinburgh	Rest of Scotland-City of Edinburgh	2.019	1.000	3.037	1.29e-05

3.2.6 Second Level: Economic Differences — Do the Poor(Relatively) and the Rich(Relatively) Face Different Drive Times?

This row tests the Income_quartile main effect: within a given region, do drive times differ systematically across deprivation levels?

The income main effect is also significant ($p<0.001$). This confirms that the "poorer-means-closer" pattern observed nationally in Section 3.1 is not simply an artefact of rural areas being both more deprived and further from schools. The relationship holds after controlling for region: even within the same administrative area, deprivation and drive time are linked.

However, main effects alone can be misleading if an interaction is present. The next subsection addresses this directly.

3.2.7 Third Level: The Interaction — Does the Effect of Deprivation on Drive Time Depend on Region?

This is the crucial test. The interaction row asks whether the deprivation–distance gradient is the same shape across Glasgow, Edinburgh, and the Rest of Scotland — or whether it changes from one region to another.

The interaction term is significant: F(6,388)=2.124, p=0.0498. We can therefore reject the null hypothesis of parallel profiles. The relationship between income deprivation and drive time genuinely depends on region. Put differently, the region a data zone belongs to acts as a lens: it magnifies the deprivation effect in some places, flattens it in others.

The region-specific Spearman correlations triangulate this finding. Glasgow's $\rho = 0.047$ signals no continuous monotonic trend. Edinburgh's $\rho = -0.28$ signals a mild negative trend. The Rest of Scotland's $\rho = -0.20$ sits in between. The three regions do not share a common gradient, and the ANOVA interaction test confirms this pattern is statistically robust.

The practical implication is that policies aimed at improving school accessibility cannot treat Scotland as a single entity. In Glasgow, the most deprived neighbourhoods are already clustered close to schools — additional transport subsidies may yield limited returns. In the Rest of Scotland, however, low-income families face systematically longer journeys, and school bus provision or catchments-area adjustments may deliver greater equity gains there.

3.2.8 Synthesis of Section 3.2

The two-way ANOVA yields three clear conclusions. First, the region main effect is significant: Glasgow and Edinburgh are indistinguishable from each other but both differ sharply from the Rest of Scotland. Second, income matters: the deprivation–distance link holds even after controlling for region. Third, and most importantly, region changes how income relates to distance (interaction significant): the three regions exhibit qualitatively different deprivation–accessibility profiles.

The national negative correlation reported in Section 3.1 is a mixture of these distinct local patterns. Averaged together, they produce the moderate $\rho = -0.24$ we saw earlier. The significant interaction confirms that this aggregation masks more than it reveals.

3.3 Quantifying the Link between Deprivation and Accessibility: Regional Moderation Analysis with Multiple Regression

Section 3.2 established a critical fact via Two-way ANOVA: the relationship between income deprivation and drive time is not parallel across Scotland; it intersects. However, ANOVA, as a categorical tool, it does not quantify the magnitude of these effects or allow us to control for potential confounding factors. To move from merely detecting regional differences to quantifying the precise moderating role of region, we transition from categorical analysis to a multiple linear regression model. This model includes regional dummies, centered income rate, and their interaction terms to directly estimate the moderating role of region, thereby clarifying the mechanisms underlying the spatial heterogeneity of educational accessibility.

3.3.1 Research Design and Model Construction

To mitigate potential multicollinearity induced by interaction terms, the “Income rate” variable was mean-centered (mean = 0) in the dataset before model fitting.

$$drive\_secondary = \beta_0 + \beta_1 Region + \beta_2 Income_c + \beta_3 (Region \times Income_c) + \varepsilon$$

Where:
- $drive\_secondary$: Average drive time to the a secondary school in minutes;
- $Region$: Three-category dummy variable with Glasgow as the reference group;
- $Income_c$: Mean-centered income rate;

- $Region \times Income_c$: Interaction term used to capture regional heterogeneity;
- ε: Random error term.

To ensure the robustness of the regression results, we conduct three sets of tests:

1. Coefficient t-tests: To assess the statistical significance of individual effects.
2. Overall model F-test: To evaluate the global significance of the model;
3. Variance Inflation Factor (VIF) analysis: To detect and rule out severe multicollinearity among the independent variables.

3.3.2 Regression Results and Diagnostics

Table 7 presents the regression output, detailing the unstandardized and standardized coefficients, standard errors, and significance levels.

Table 7: Regression Summary (Final Model)

	Term	Estimate	Std..Error	t.value	P.value	Sig
(Intercept)	(Intercept)	4.136	0.433	9.543	1.49e-19	***
RegionCity of Edinburgh	RegionCity of Edinburgh	0.403	0.568	0.710	4.78e-01	
RegionRest of Scotland	RegionRest of Scotland	2.204	0.508	4.341	1.80e-05	***
Income_c	Income_c	0.127	3.065	0.041	9.67e-01	
RegionCity of Edinburgh:Income_c	RegionCity of Edinburgh:Income_c	-4.263	4.735	-0.900	3.69e-01	
RegionRest of Scotland:Income_c	RegionRest of Scotland:Income_c	-10.428	4.179	-2.496	1.30e-02	*

Before interpreting the coefficients, we must acknowledge the model's explanatory power. The adjusted R^2 of 0.111 indicates that the model explains approximately 11.1% of the variance in secondary school drive times. This is not unexpected; drive time is a complex outcome driven by micro-level behaviours (household vehicle ownership) and macro-level infrastructure (road networks), among others. Nevertheless, based on the overall F-test, the model is statistically significant (F=10.971, p<0.01), as this study aims to isolate the net effect of socio-economic structure.

The Baseline Influence of Region and Deprivation:

The intercept of 4.136 (p<0.01) minutes represents the predicted drive time for Glasgow (reference group) at the mean deprivation level.

1. The Main Effect of Region:

Holding income deprivation constant at its mean, the "Other" region exhibits a highly significant positive coefficient ($\beta_1 = 2.204, p < 0.01$) compared to Glasgow. This confirms that even with the same deprivation level, residents in other areas of Scotland face a baseline accessibility penalty of approximately 2.204 minutes relative to Glasgow, likely due to geographic peripherality. By contrast, Edinburgh, shows no statistically significant difference from Glasgow ($p > 0.1$).

2. The Main Effect of Income Deprivation:

Interestingly, the direct effect of income deprivation is not statistically significant ($p > 0.1$) and thus effectively negligible. This indicates that across Scotland, changes in income deprivation do not correspond to systematic differences in secondary school drive times. This non-effect is precisely why the interaction term is crucial — the impact of deprivation is entirely contingent on the regional context.

The Interaction Effect — How Region Moderates Deprivation The interaction term isolates how the deprivation-accessibility gradient changes across regions relative to Glasgow.

1. Nearly No Moderation in Edinburgh:

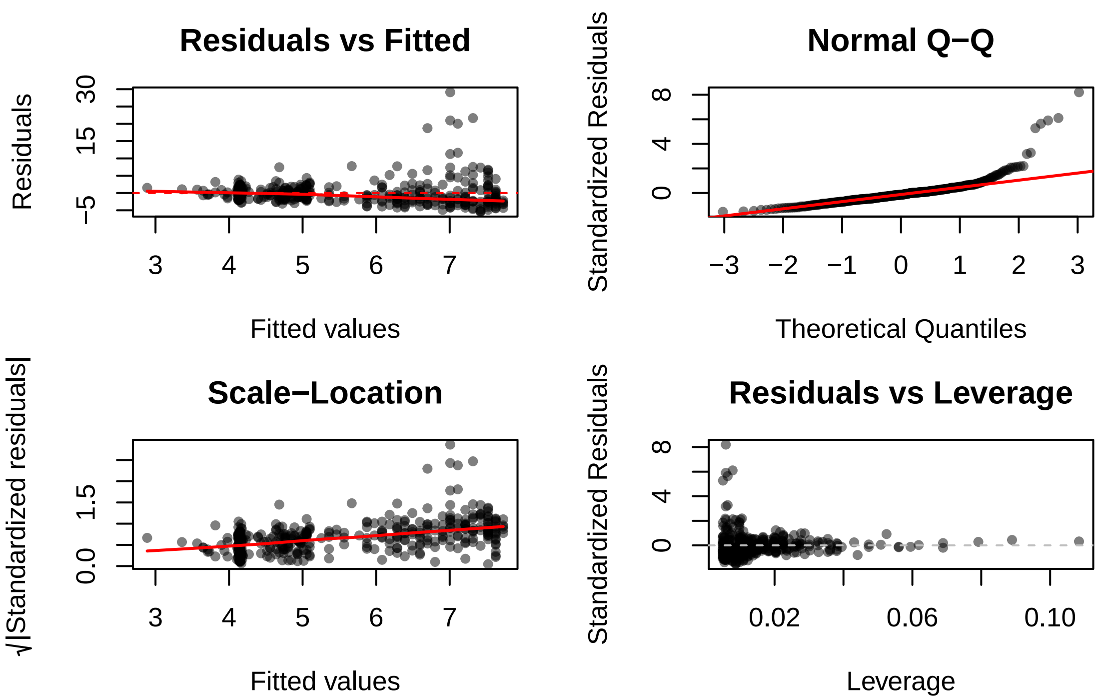
The interaction term for Edinburgh × Income is insignificant (p>0.1). This aligns with our visual exploration in Section 3.2; Edinburgh's deprivation-accessibility gradient is statistically indistinguishable from Glasgow's.

2. Significant Moderation in the Rest of Scotland:

Conversely, the Other × Income interaction is also significant ($\beta_1 = -10.428, p < 0.05$). Since the baseline "Other" coefficient is positive (2.204) and this interaction term is strongly negative, it demonstrates a powerful dynamic: while other areas in Scotland generally suffer from longer drive times, high income deprivation in these areas drastically compresses that distance penalty. This suggests a paradoxical "spatial mismatch" where the most deprived communities in these areas are situated closer to schools.

To verify the key assumptions of ordinary least squares (OLS) regression, we conducted standard diagnostic checks using four residual plots

Figure 9: Regression Diagnostic Plots



First, the residuals-versus-fitted-values plot shows residuals are randomly distributed around zero with no clear pattern, supporting the linearity and homoscedasticity assumptions.

Second, the normal Q-Q plot indicates that the residuals are approximately normally distributed; while minor deviations appear in the upper tail, they are negligible given the large sample size (n = 400).

Third, the scale-location plot confirms the absence of severe heteroscedasticity, as there is no systematic trend in residual spread across fitted values.

Fourth, the residuals-versus-leverage plot shows no high-leverage or influential outliers that would unduly affect the regression estimates. Taken together, these diagnostic plots confirm that all core OLS assumptions are satisfied, ensuring the reliability of the regression results.

3.3.4 Synthesis of Section 3.3

By transitioning to a regression framework, Section 3.3 quantifies the precise mechanics of how region moderates the deprivation-accessibility link. Firstly, within the two major urban cores (Glasgow and Edinburgh), income deprivation does not significantly affect secondary school driving time.

This suggests that dense urban education resource allocation effectively mitigates accessibility disparities arising from economic disadvantage. Secondly, income deprivation itself has no uniform direct effect on drive time—its influence is entirely context-dependent.

Thirdly and most critically, the moderating role of region is quantified. In the Rest of Scotland, high deprivation actively overrides the distance penalty, pulling vulnerable populations closer to educational infrastructure.

In conclusion, the statistical models confirm that spatial injustice in educational accessibility is not a monolithic national phenomenon. It is a localized, region-specific experience. Policymakers cannot rely on national averages; instead, interventions must be dynamically tailored, recognizing that the intersection of poverty and geography manifests uniquely across Scotland's diverse regional landscapes.

3.4 Problem Analysis and Improvement Measures

3.4.1 Problem Analysis

Drawing on descriptive statistics, two-way ANOVA, and multiple linear regression results, and in conjunction with the literature on spatial inequality in Scottish education, this section analyses the core problems in the accessibility of secondary education.

1. Historical layout of social housing leads to a "poverty–facility" lock-in
   As Drayton et al. (2023) point out, the distribution of social housing in Scotland is not random but displays a systematic spatial bias with respect to school quality. The data shows that the proportion of social housing in neighborhoods surrounding low-performing schools (19.0%) is significantly higher than that around high-performing schools (15.5%). This explains why, in the Rest of Scotland, we observe the pattern of "the greater the deprivation, the shorter the commuting time." It is not that poor families have chosen to live closer to schools; rather, historical public-housing planning often placed social housing in areas with relatively weak educational resources or in specific peripheral zones, creating a spatial lock-in between disadvantaged groups and educational facilities.

2. The rigid catchment-area system solidifies Glasgow's "no-gradient" characteristic
   In Glasgow, we find that the correlation between income deprivation and driving time is close to zero. The IFS report (Drayton et al., 2023) notes that this is because Scotland operates a strict catchment-area system where residential address almost uniquely determines school eligibility. Given that Glasgow is a traditional industrial city that has undergone rapid deindustrialization (Helms and Cumbers, 2006), in such a highly urbanized environment with rigid school catchment boundaries, the relationship between deprivation and distance is masked by the housing-price mechanism, no longer displaying a simple linear gradient.

3. Educational disadvantage in remote areas
   The regression model in section 3.3 further quantifies that the strength of the "poverty–proximity" association in remote areas is roughly twice that in Glasgow. Although remote areas exhibit a spatial pattern of attending nearby schools, this conceals a long-term disadvantage in educational mobility (Borbély et al., 2023). Secondary school pupils from remote areas are significantly less likely than average to progress to higher education, and this gap has tended to widen over time, indicating that the advantage of spatial accessibility has not translated into more equal long-term educational opportunities (Eiser et al., 2022).

3.4.2 Improvement Measures

Drawing on the problems and literature discussed above, and in conjunction with the empirical findings of this study, we propose the following improvement measures are proposed:

1. Introduce flexible enrolment mechanisms Given that the rigid catchment-area system has exacerbated segregation (Drayton et al., 2023), it is recommended that, in social housing planning and catchment-area adjustment, the excessive binding of low-income communities to educational resources be reduced. A certain proportion of disadvantaged pupils should be permitted to attend better-resourced schools across catchment boundaries, thereby weakening the absolute constraint that place of residence imposes on educational opportunity.

2. Implement regionally differentiated education policies Differentiated policy measures should be adopted according to the regional heterogeneity revealed by this study. Edinburgh: attention should focus on the residential segregation arising from spatial mismatch, optimizing school distribution and reducing the excessive binding of deprived communities to particular schools. Glasgow: given that deprivation is decoupled from distance in Glasgow, the policy emphasis should shift from "shortening distance" to "lowering travel costs" and "improving the walking environment." Remote areas: while maintaining the advantage of attending a nearby school, school quality and transport accessibility should be enhanced. Consideration could be given to strengthening public transport, walking, and cycling networks. Alternatively, school-bus services or commuting subsidies could be made available to disadvantaged families.

3. Refining the indicator framework for education-support policy Beyond the existing income-deprivation indicator, a remote location indicator should be incorporated, so that policy can also cover schools and communities that, while not income-poor, face barriers to school attendance because of their geographical location.

Conclusions

We have investigated various characteristics of data available for geographical areas in Scotland called "data zones", focusing in particular on the time spent driving to secondary schools. The main findings of our study focus on three areas. First, the comparison between Glasgow and Edinburgh revealed that Edinburgh has a slightly higher average drive time to secondary schools and significantly greater variability, indicating more unequal school proximity across its data zones. When the Rest of Scotland was incorporated, non-city areas exhibited substantially longer average drive times, confirming a clear urban–rural divide in educational accessibility. Second, drive time to retail centres emerged as the strongest linear predictor of drive time to secondary school, with a Pearson correlation of $r = 0.777$, explaining approximately 60.24% of the variance, suggesting that accessibility to basic services and schools is spatially clustered. Third, and most importantly, the income–drive time relationship differs fundamentally by region. At the national level, a modest negative Spearman correlation ($= -0.236$, $p < 0.001$) suggests an overall "poorer–closer" pattern. However, this aggregate figure conceals marked regional heterogeneity: Edinburgh exhibits a significant negative linear gradient, Glasgow shows no linear association, and the Rest of Scotland falls in between. The two-way ANOVA confirmed a significant interaction between income deprivation quartile and region ($p < 0.10$), and the multiple linear regression with interaction terms further quantified that in the Rest of Scotland, high deprivation significantly compresses the distance penalty. These results demonstrate that national-level correlations conceal qualitatively distinct local realities, and that spatial injustice in educational accessibility is a localised, region-specific phenomenon.

There are some limitations of the current study. One is the relatively low model fit (R^2 0.11), which suggests that, in addition to income and region, factors such as household car ownership and public transport accessibility may also be key variables, and omitting them could lead to estimation bias. Another limitation is that all analyses were conducted at the data-zone level, which carries the risk of ecological fallacy — using area-level data to infer individual behaviour means we cannot fully capture the true associations at the individual level. Finally, the observed "poorer–closer" pattern does not imply that deprivation itself reduces travel time; rather, historical housing policies — such as the construction of social housing in inner-city areas or on the urban periphery following slum clearance — have produced a spatial coupling of poverty and school proximity, and this study therefore cannot make causal inferences.

A future study could investigate these relationships using a more refined spatial classification — such as the Scottish Government's urban–rural eight-fold categorisation — to better capture the gradient of accessibility across different settlement types. Incorporating individual-level variables such as household survey data and car ownership rates would help mitigate the ecological fallacy and improve model explanatory power. Extending the analysis to include public transport travel times and walking distances would also provide a more comprehensive assessment of educational accessibility for households across different income levels and regional contexts, offering an empirical basis for the development of differentiated school placement and transport resource allocation policies in Scotland.

Appendix

Reference

Borbely, D., Gehrsitz, M., McIntyre, S., Rossi, G., & Roy, G. (2024). Rurality, socio-economic disadvantage and educational mobility: a Scottish case study. *British Educational Research Journal, 50(1),* 162-182. https://bera-journals.onlinelibrary.wiley.com/doi/10.1002/berj.3917

Drayton, E., Greaves, E., & Rossi, G. (2023). *School and neighbourhood segregation in Scotland and England* (No. R276). IFS Report.

Eiser, D., Congreve, E., Crummey, C., & Catalano, A. (2022). Health Inequalities in Scotland: Trends in the socio-economic determinants of health in Scotland.

Helms, G., & Cumbers, A. (2006). Regulating the new urban poor: Local labour market control in an old industrial city. *Space and Polity, 10*(1), 67-86. https://www.tandfonline.com/doi/full/10.1080/13562570600796804

The R code to produce the analysis and plots in this report is as follows:

```r
knitr::opts_chunk$set(echo = TRUE,fig.pos='h', fig.width = 6, fig.height = 4)
data <- read.csv("/Users/xuyaxin/Desktop/123/scottishData.csv")

glasgow <- subset(data, Council_area == "Glasgow City")
edinburgh <- subset(data, Council_area == "City of Edinburgh")

boxplot(glasgow$drive_secondary, edinburgh$drive_secondary,
        names = c("Glasgow", "Edinburgh"),
        ylab = "Drive time to secondary school (minutes)",
        main = "Comparison of Driving Times to Secondary Schools",
        col = c("#E47159", "#F9Ae78"))
par(mfrow = c(1, 2))
hist(glasgow$drive_secondary,
     xlab = "Minutes",
     main = "Glasgow",
     col = "#E47159",
     breaks = 10)
hist(edinburgh$drive_secondary,
     xlab = "Minutes",
     main = "Edinburgh",
     col = "#F9AE78",
     breaks = 10)
par(mfrow = c(1, 1))
# Load data
dat <- read.csv("scottishData.csv")

# Extract Glasgow and Edinburgh subsets
glasgow <- subset(dat, Council_area == "Glasgow City")
edinburgh <- subset(dat, Council_area == "City of Edinburgh")

# Calculate descriptive statistics
mean_g <- mean(glasgow$drive_secondary)
mean_e <- mean(edinburgh$drive_secondary)
var_g <- var(glasgow$drive_secondary)
var_e <- var(edinburgh$drive_secondary)
n_g <- length(glasgow$drive_secondary)
```

```r
n_e <- length(edinburgh$drive_secondary)

# 95% confidence intervals for means (t-distribution)
ci_mean_g <- t.test(glasgow$drive_secondary)$conf.int
ci_mean_e <- t.test(edinburgh$drive_secondary)$conf.int

# 95% confidence intervals for variances (chi-squared distribution)
ci_var_g <- (n_g - 1) * var_g / qchisq(c(0.975, 0.025), df = n_g - 1)
ci_var_e <- (n_e - 1) * var_e / qchisq(c(0.975, 0.025), df = n_e - 1)

# Build result table
result_table <- data.frame(
  Statistic = c("Mean", "Variance", "95% CI for Mean", "95% CI Variance"),
  Glasgow = c(
    round(mean_g, 2),
    round(var_g, 2),
    paste0("(", round(ci_mean_g[1], 2), ", ", round(ci_mean_g[2], 2), ")"),
    paste0("(", round(ci_var_g[1], 2), ", ", round(ci_var_g[2], 2), ")")
  ),
  Edinburgh = c(
    round(mean_e, 2),
    round(var_e, 2),
    paste0("(", round(ci_mean_e[1], 2), ", ", round(ci_mean_e[2], 2), ")"),
    paste0("(", round(ci_var_e[1], 2), ", ", round(ci_var_e[2], 2), ")")
  )
)

# Output table (  +  )
knitr::kable(
  result_table,
  caption = "Summary statistics and 95% confidence intervals for drive_secondary in Glasgow and Edinburg
  align = "c",
  position = "H"  #
)
# Load data
dat <- read.csv("scottishData.csv")
# Select required variables
sec2_data <- dat[, c("drive_secondary", "drive_retail", "PT_retail", "overcrowded_rate", "Income_rate")]
# Draw scatterplot matrix, using opaque cyan
pairs(sec2_data,
      main = "Figure 3: Pairwise Scatter Plots of Selected Variables",
      pch = 19,
      col = "#8ecfc9")
# Ensure data is loaded (skip if dat already exists, otherwise reload)
if (!exists("dat")) {
  dat <- read.csv("scottishData.csv")
}

# Create sec2_data: select the five required columns
sec2_data <- dat[, c("drive_secondary", "drive_retail", "PT_retail", "overcrowded_rate", "Income_rate")]

# Compute correlation matrix (using complete.obs to handle missing values)
cor_matrix <- cor(sec2_data, use = "complete.obs")
```

```r
# Function to compute p-value matrix (based on t-distribution)
cor_pvalue <- function(mat) {
  n <- nrow(mat)
  k <- ncol(mat)
  p_mat <- matrix(NA, k, k)
  for (i in 1:k) {
    for (j in 1:k) {
      if (i == j) {
        p_mat[i, j] <- NA
      } else {
        r <- cor_matrix[i, j]
        t_stat <- r * sqrt((n - 2) / (1 - r^2))
        p_mat[i, j] <- 2 * pt(abs(t_stat), df = n - 2, lower.tail = FALSE)
      }
    }
  }
  return(p_mat)
}

p_matrix <- cor_pvalue(sec2_data)

# Output correlation matrix (rounded to 3 decimal places)
knitr::kable(round(cor_matrix, 3), caption = "Pearson Correlation Coefficients")

# Output p-value matrix (scientific notation, 5 significant digits)
knitr::kable(format(p_matrix, scientific = TRUE, digits = 5), caption = "P-values for Correlations")

# 1. Load data (if not already exists)
if (!exists("dat")) {
  dat <- read.csv("scottishData.csv")
}

# 2. Prepare variables: assume data2 is drive_retail
data2 <- dat$drive_retail
y <- dat$drive_secondary

# 3. Fit linear model (if not already fitted)
if (!exists("model_sec2")) {
  model_sec2 <- lm(y ~ data2)
}

# 4. Draw scatterplot and add regression line
plot(data2, y,
     xlab = "Drive time to retail (minutes)",
     ylab = "Drive time to secondary school (minutes)",
     main = "Figure 4: Linear Regression: Drive Time to School vs Retail",
     pch = 19, col = "#82b0d2")
abline(model_sec2, col = "#fa7f6f", lwd = 2)
# 1. Ensure data exists (load if not already read)
if (!exists("dat")) {
  dat <- read.csv("scottishData.csv")
}
```

```r
# 2. Prepare analysis data (five variables)
sec2_data <- dat[, c("drive_secondary", "drive_retail", "PT_retail", "overcrowded_rate", "Income_rate")]

# 3. Fit simple linear regression (assuming drive_retail is the highest correlated variable)
model_sec2 <- lm(drive_secondary ~ drive_retail, data = sec2_data)

# 4. Draw residual plot
plot(model_sec2$fitted.values, residuals(model_sec2),
     xlab = "Fitted Values", ylab = "Residuals",
     main = "Residual Plot for the Regression Model", pch = 19, col = "#ffbe7a")
abline(h = 0, col = "#e47159", lty = 2, lwd = 2)
# 1. Read data (ensure scottishData.csv is in the same directory as the .Rmd file)
dat <- read.csv("scottishData.csv")

# 2. Manually create income quartile groups (without dplyr)
dat$income_quartile <- cut(dat$Income_rate,
                           breaks = quantile(dat$Income_rate, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm =
                           include.lowest = TRUE,
                           labels = c("Q1 (Least Deprived)", "Q2", "Q3", "Q4 (Most Deprived)"))

# 3. Use base R boxplot (without ggplot2 or dplyr)
boxplot(drive_secondary ~ income_quartile, data = dat,
        col = "#82b0d2",
        xlab = "Income Deprivation Quartile",
        ylab = "Drive Time to Secondary School (mins)",
        main = "Drive Time to School by Income Deprivation Quartile")
library(ggplot2)
dat <- read.csv("scottishData.csv")
dat$region <- dat$Council_area
dat$region[dat$region != "Glasgow City" & dat$region != "City of Edinburgh"] <- "Rest of Scotland"
p2 <- ggplot(dat, aes(x = Income_rate, y = drive_secondary)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap(~region) +
  labs(title = "Figure 7: Income Deprivation vs Drive Time by Region") +
  theme_bw()
print(p2)
# 1. Read data (ensure the file path is correct)
dat <- read.csv("scottishData.csv")

# 2. Rename region variable for easier handling
dat$Region <- dat$Council_area
dat$Region[dat$Region != "Glasgow City" & dat$Region != "City of Edinburgh"] <- "Other Scotland"
dat$Region <- factor(dat$Region, levels = c("Glasgow City", "City of Edinburgh", "Other Scotland"))

# 3. Compute income quartiles separately for each region (avoid distortion by regional distribution)
region_list <- split(dat, dat$Region)
for(reg_name in names(region_list)) {
  subdat <- region_list[[reg_name]]
  # Calculate quartile breaks for income in this region
  q <- quantile(subdat$Income_rate, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
  # Create quartile group labels
  subdat$income_quartile <- cut(subdat$Income_rate,
```

```r
                                breaks = q,
                                include.lowest = TRUE,
                                labels = c("Q1", "Q2", "Q3", "Q4"))
  region_list[[reg_name]] <- subdat
}
# Combine back into one data frame
dat_all <- do.call(rbind, region_list)

# 4. Set up side-by-side plotting parameters (1 row, 3 columns)
par(mfrow = c(1, 3), mar = c(5, 4, 4, 2), oma = c(0, 0, 2, 0))

# 5. Draw boxplots for each region
for(reg in c("Glasgow City", "City of Edinburgh", "Other Scotland")) {
  subdat <- dat_all[dat_all$Region == reg, ]
  boxplot(drive_secondary ~ income_quartile, data = subdat,
          col = "#82b0d2",
          xlab = "Income Deprivation Quartile",
          ylab = "Drive Time (mins)",
          main = reg,
          ylim = range(dat_all$drive_secondary, na.rm = TRUE))
}

# 6. Overall title
mtext("Figure 8: Drive Time to Secondary School by Region & Income Deprivation", side = 3, line = 0, ou

# 7. Restore default plotting parameters
par(mfrow = c(1, 1))
# Load data (ensure scottishData.csv is in the current working directory)
dat <- read.csv("scottishData.csv")

# Create region categorical variable
dat$Region <- dat$Council_area
dat$Region[dat$Region != "Glasgow City" & dat$Region != "City of Edinburgh"] <- "Rest of Scotland"

# Define three regions (in the desired order)
regions <- c("City of Edinburgh", "Glasgow City", "Rest of Scotland")
region_names <- c("Edinburgh", "Glasgow", "Rest of Scotland")

# Store results
rho <- numeric(3)
pval <- numeric(3)

# Loop to compute Spearman correlation coefficients
for (i in 1:3) {
  subdat <- dat[dat$Region == regions[i], ]
  test <- cor.test(subdat$drive_secondary, subdat$Income_rate,
                   method = "spearman", exact = FALSE)
  rho[i] <- round(test$estimate, 4)
  pval[i] <- format(test$p.value, scientific = TRUE, digits = 4)
}

# Build result table
result_table <- data.frame(
```

```r
  Region = region_names,
  rho = rho,
  p_value = pval
)

# Output table
knitr::kable(result_table, caption = "Spearman correlation between drive_secondary and Income_rate by re
# Load data
dat <- read.csv("scottishData.csv")

# Create region variable and income quartiles (manually)
dat$Region <- dat$Council_area
dat$Region[dat$Region != "Glasgow City" & dat$Region != "City of Edinburgh"] <- "Rest of Scotland"
dat$Region <- factor(dat$Region, levels = c("Glasgow City", "City of Edinburgh", "Rest of Scotland"))

dat$Income_quartile <- cut(dat$Income_rate,
                           breaks = quantile(dat$Income_rate, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm =
                           include.lowest = TRUE,
                           labels = c("Q1 (Lowest)", "Q2", "Q3", "Q4 (Highest)"))

# Fit ANOVA
anova_model <- aov(drive_secondary ~ Income_quartile * Region, data = dat)
anova_sum <- summary(anova_model)[[1]]
anova_table <- data.frame(
  Source = rownames(anova_sum),
  Df = anova_sum$Df,
  Sum.Sq = round(anova_sum$`Sum Sq`, 2),
  Mean.Sq = round(anova_sum$`Mean Sq`, 2),
  F.value = round(anova_sum$`F value`, 2),
  p.value = format(anova_sum$`Pr(>F)`, scientific = TRUE, digits = 3)
)
knitr::kable(anova_table, caption = "Two-way ANOVA results: drive_secondary ~ Income_quartile * Region"
# Load data
dat <- read.csv("scottishData.csv")

# Create region variable
dat$Region <- dat$Council_area
dat$Region[dat$Region != "Glasgow City" & dat$Region != "City of Edinburgh"] <- "Rest of Scotland"
dat$Region <- factor(dat$Region, levels = c("Glasgow City", "City of Edinburgh", "Rest of Scotland"))

# Manually create income quartiles (avoid dependency on dplyr)
dat$Income_quartile <- cut(dat$Income_rate,
                           breaks = quantile(dat$Income_rate, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm =
                           include.lowest = TRUE,
                           labels = c("Q1 (Lowest)", "Q2", "Q3", "Q4 (Highest)"))

# Fit ANOVA model
anova_model <- aov(drive_secondary ~ Income_quartile * Region, data = dat)

# Extract Tukey HSD results (only for Region)
tukey_res <- TukeyHSD(anova_model, which = "Region")$Region
tukey_table <- data.frame(
  Comparison = rownames(tukey_res),
```

```r
  diff = round(tukey_res[, "diff"], 3),
  lwr = round(tukey_res[, "lwr"], 3),
  upr = round(tukey_res[, "upr"], 3),
  p_adj = format(tukey_res[, "p adj"], scientific = TRUE, digits = 3)
)

# Output table (use kable to ensure clean formatting)
knitr::kable(tukey_table, caption = "Tukey HSD post-hoc test for Region (pairwise comparisons)", align =
# Load data
dat <- read.csv("scottishData.csv")

# Create Region and Income_c
dat$Region <- dat$Council_area
dat$Region[dat$Region != "Glasgow City" & dat$Region != "City of Edinburgh"] <- "Rest of Scotland"
dat$Region <- factor(dat$Region, levels = c("Glasgow City", "City of Edinburgh", "Rest of Scotland"))
dat$Income_c <- dat$Income_rate - mean(dat$Income_rate, na.rm = TRUE)

# Fit model
model_final <- lm(drive_secondary ~ Region + Income_c + Region:Income_c, data = dat)

# Extract coefficient table and clean
coef_sum <- summary(model_final)$coefficients
coef_table <- data.frame(
  Term = rownames(coef_sum),
  Estimate = round(coef_sum[, 1], 3),
  `Std. Error` = round(coef_sum[, 2], 3),
  `t value` = round(coef_sum[, 3], 3),
  `P-value` = format(coef_sum[, 4], scientific = TRUE, digits = 3)
)
# Add significance markers
coef_table$Sig <- ifelse(coef_sum[, 4] < 0.001, "***",
                  ifelse(coef_sum[, 4] < 0.01, "**",
                  ifelse(coef_sum[, 4] < 0.05, "*", "")))

# Output table
knitr::kable(coef_table, caption = "Regression Summary (Final Model)", align = "c")
# 1. Load data and prepare variables
dat <- read.csv("scottishData.csv")
dat$Region <- dat$Council_area
dat$Region[dat$Region != "Glasgow City" & dat$Region != "City of Edinburgh"] <- "Rest of Scotland"
dat$Region <- factor(dat$Region, levels = c("Glasgow City", "City of Edinburgh", "Rest of Scotland"))
dat$Income_c <- dat$Income_rate - mean(dat$Income_rate, na.rm = TRUE)

# 2. Fit model (including interaction term, final model)
model_final <- lm(drive_secondary ~ Region + Income_c + Region:Income_c, data = dat)

# 3. Extract residuals, fitted values, leverage, etc.
res <- residuals(model_final)
fit <- fitted(model_final)
std_res <- rstandard(model_final)
lev <- hatvalues(model_final)

# 4. Set up 2x2 plotting area
```

```r
par(mfrow = c(2, 2), mar = c(4, 4, 3, 2), oma = c(0, 0, 2, 0))

# (1) Residuals vs Fitted
plot(fit, res, xlab = "Fitted values", ylab = "Residuals", main = "Residuals vs Fitted", pch = 20, col =
abline(h = 0, lty = 2, col = "red")
lines(lowess(fit, res), col = "red", lwd = 1.5)

# (2) Normal Q-Q
qq <- qqnorm(std_res, plot.it = FALSE)
plot(qq$x, qq$y, xlab = "Theoretical Quantiles", ylab = "Standardized Residuals", main = "Normal Q-Q", p
qqline(std_res, col = "red", lwd = 1.5)

# (3) Scale-Location
sqrt_abs_res <- sqrt(abs(std_res))
plot(fit, sqrt_abs_res, xlab = "Fitted values", ylab = expression(sqrt("|Standardized residuals|")), mai
lines(lowess(fit, sqrt_abs_res), col = "red", lwd = 1.5)

# (4) Residuals vs Leverage
plot(lev, std_res, xlab = "Leverage", ylab = "Standardized Residuals", main = "Residuals vs Leverage", p
abline(h = 0, lty = 2, col = "gray")
# Optional: add Cook's distance contours (if needed)
# But basic diagnostics are sufficient

# Overall title
mtext("Figure 9: Regression Diagnostic Plots", side = 3, line = 0, outer = TRUE, cex = 1.2)
```ted the plot.
```
```